

PROJECT PHASE-II • REVIEW-I

EDUCloud

Connected on One Platform.

Secure Cloud Student Information Management System with Privacy Protection

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Problem Statement

- 1 Institutions still rely on manual or fragmented digital systems for student records, causing redundancy, inconsistency and administrative overload.
- 2 Cloud adoption improves accessibility and scale — but most existing cloud SIS platforms lack strong authentication, RBAC and encryption.
- 3 Sensitive academic data needs confidentiality, integrity and controlled access without sacrificing real-time usability.

Updated Abstract

EDUCloud is a cloud-based Student Information Management System that centralizes student, faculty and admin workflows on one secure platform. Building on Phase-I's analysis and design, Phase-II moves into implementation: **the React.js frontend for all three user roles is now functional**, while the Java Spring Boot backend, MySQL database, RBAC enforcement and AES/JWT security layer are under active development on AWS infrastructure. The system targets real-time attendance, results and notification workflows, with encryption, hashing and secure sessions protecting data at rest and in transit.

Cloud-Based System Architecture

Three-tier design

Presentation

React.js frontend on S3 + CloudFront — dashboards, attendance, results, AI chatbot

Application

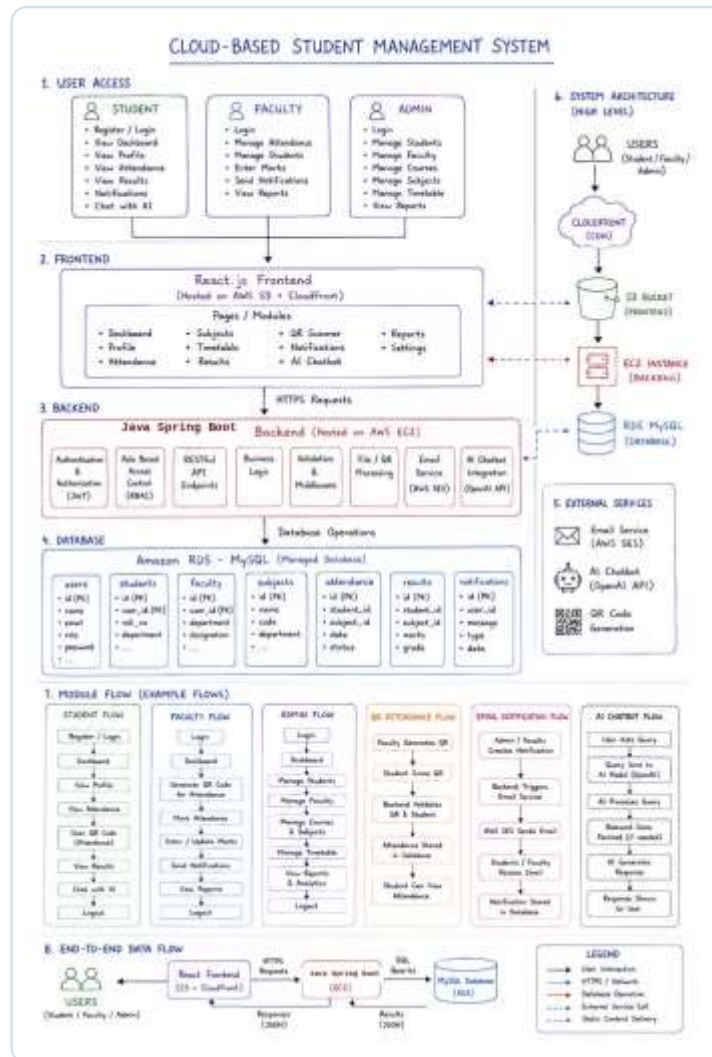
Java Spring Boot on EC2 — JWT auth, RBAC, REST APIs, validation, email & QR services

Data

Amazon RDS (MySQL) — normalized tables for users, students, faculty, attendance, results

Deployment

AWS: CloudFront (CDN) → S3 (frontend) → EC2 (backend) → RDS (MySQL), with SES for email and OpenAI API for the chatbot.



Role-Based Modules

Student

- Register / Login
- Dashboard & profile
- View attendance
- QR attendance check-in
- View results
- Notifications
- AI chatbot

Faculty

- Login & dashboard
- Manage attendance
- Generate attendance QR
- Manage students
- Enter marks
- Send notifications
- View reports

Admin

- Login & dashboard
- Manage students & faculty
- Manage courses / subjects
- Manage timetable
- View reports

Implementation Status

FRONTEND — COMPLETED

- React.js application built for all three roles
- Dashboard, profile, attendance & results views
- Faculty attendance & marks-entry screens
- Admin management screens (students, faculty, courses, timetable)
- QR scanner, notifications and AI chatbot UI wired up

BACKEND & DATA — IN PROGRESS

- Java Spring Boot REST API endpoints
- MySQL schema on Amazon RDS
- JWT authentication & RBAC middleware
- AES encryption + bcrypt password hashing
- AWS deployment (EC2, S3, CloudFront, SES)

Security & Privacy Design

1

JWT Authentication

Stateless, verified sessions for every request

2

Role-Based Access

Student / Faculty / Admin scoped permissions

3

AES Encryption

Sensitive data encrypted at rest

4

bcrypt / SHA-256

One-way password hashing — no plaintext storage

5

HTTPS / SSL-TLS

Encrypted data in transit

6

Audit Logging (planned)

Access & activity tracking for accountability

Testing & Validation Plan

Functional Testing

Frontend — done

Verify navigation, forms & role dashboards render and behave correctly

Unit / API Testing

Backend — processing

Validate REST endpoints, request/response contracts and error handling

Security Testing

Processing

Verify JWT/RBAC enforcement, AES encryption and injection resistance

Load Testing

Processing

Measure response time and throughput under concurrent user access

Data Integrity Testing

Processing

Confirm consistency of records after create/update/delete operations

Performance Evaluation Criteria

To be measured once the backend and database are integrated (Review-II):

1

Response Time

API latency under typical and peak load

2

Scalability

Behavior under increasing concurrent users

3

Access Correctness

RBAC boundaries hold across all roles

4

Data Consistency

Accuracy after synchronized multi-user updates

5

Security Robustness

Resistance to unauthorized access attempts

6

Availability

Uptime and recovery under AWS auto-scaling

Existing Systems vs. EDUCloud

Criteria	Existing Systems	EDUCloud (Proposed)
Data storage	Manual / isolated files	Centralized on AWS cloud
Access control	Basic login only	JWT + role-based (RBAC)
Encryption	Little or none	AES + HTTPS/SSL
Scalability	Limited, on-premise	Auto-scaling, load-balanced
Real-time sync	Rare, delayed	Instant, cross-role
Monitoring & audit	Minimal or absent	Access logs & activity tracking
Attendance capture	Manual roll-call	QR-based, instant marking

Documentation & Publication Status

Documentation Progress

- Phase-I report complete: intro, literature survey, methodology
- System architecture & module-flow diagrams finalized
- Database schema (ER-level) documented
- Frontend module documentation in progress
- Final report / thesis drafting to begin after backend integration

Publication / Patent

- No publication or patent filed yet
- Target: submit a paper after performance evaluation results (post Review-III)
- Candidate venues: departmental / national student conferences

Key References

- Ubamadu et al., “Cloud-Based Higher Education Management Framework: Enhancing Student Administration in U.S. Universities.”
- Hughes, J., “Adoption of Cloud-Based Management Systems in Higher Education Institutions in Asia.”
- Chen, W., “Construction of Student Information Management System for Universities Based on Cloud Computing Platform.”
- Wandira, Karanja & Mwangi, “Cloud-Based Student Information Management Systems and Data Security Challenges in Higher Education Institutions.”
- Chinnasamy et al., “Secure Student Record Management using Visual Cryptography in Educational Systems.”
- Alameri, I. A. & Radchenko, G., “Development of Student Information Management System based on Cloud Computing Platform.”

Where We Are, and What's Next

Frontend for all three roles is functional and demonstrated. Backend, database, security layer and AWS deployment are the focus for the next milestone.

1

Review-II (this stage)

Frontend demo, 100% UI flows, documentation progress

2

Next 3-4 weeks

Build & integrate Java Spring Boot APIs with MySQL on RDS

3

Review-III

Security hardening, performance & load testing, results

4

Final Review

Complete demonstration, final report, publication draft